Department of Computer Science and Engineering

Faculty of Engineering & Technology, Gurukula Kangri Deemed to be University, Haridwar

First Sessional Exam

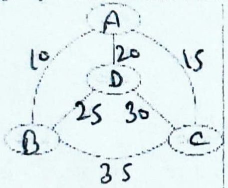
Subject: Design and Analysis of Algorithm

Paper Code: BCE-C523

Time: 1 Hr MM: 20

Note: Answer any three questions from Section A (each question carries 4 marks).

Answer any one question from Section B (each question carries 8 marks).

Q.No.	Section A (Questions of 4 marks each)	Keyword	CO	BT
Q1.	Explain asymptotic notation (Big Oh, Big Theta, and Big Omega).	Remembering	(CO1)	BT1
Q2.	The following graph shows a set of cities and distance between every pair of cities If the salesman starting city is A, then find the TSP tour in the graph? 	Applying	(CO4)	BT3
Q3.	Explain Floyd-Warshall algorithm with an example.	Understanding	(CO2)	BT2
Q4.	Given the set: {7,3,2,5,8,10,11} and target sum: S=21, Is there a subset of this set that sums exactly to 21? If yes, find one such subset.	Analyzing	(CO6)	BT4
	Section B (Questions of 8 marks each)			
Q1.	Define N-Queen Problem and find its time and space complexity.	Applying	(CO3)	BT3
Q2.	Solve any two of the following recurrence relation by using Master Method a. $T(n) = T(n/2) + 1$ b. $T(n) = 2T(n/2) + n$ c. $T(n) = 2T(n/2 + 17) + n$.	Evaluating	(CO4)	BT5

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Q.No.	Section A (Questions of 4 marks each)	Keyword	CO	BT
Q1.	Define Heap Sort using with the suitable example.	Remembering	(CO1)	BT1
Q2.	Insert the following sequence of numbers into an initially empty Red–Black Tree: 10,20,30,15,25,27	Applying	(CO4)	BT3
Q3.	Define Bellman-Ford Algorithm with the help of example.	Understanding	(CO2)	BT2
Q4.	Explain Sorting in Linear time.	Analyzing	(CO6)	BT4
	Section B (Questions of 8 marks each)			
Q1.	Explain the Matrix Chain Multiplication problem in details.	Applying	(CO3)	BT3
Q2.	Color the following graph using the three color (red, green ,blue) V={A,B,C,D,E} E={(A,B),(A,C),(B,C),(B,D),(C,D),(C,E),(D,E)}	Evaluating	(CO4)	BT5